OpticsBox

Equation and matrix reference

Every formula the simulator uses: the web app (`OpticsBox.html`) and the Python library (`opticsbox.py`) implement exactly these, and agree to about 10^{-13}.

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1 Conventions

Field vector. At any cross-section of a circuit with N waveguides the state is

$$\mathbf{E} = \left(E_1^+,\, E_1^-,\, E_2^+,\, E_2^-,\, \ldots,\, E_N^+,\, E_N^-\right)^{\mathsf{T}},$$

where $+$ travels forward (left to right, input to output), $-$ travels backward, and the index is the waveguide. In the *forward-only* model the backward entries are dropped and $\mathbf{E} = (E_1^+, \ldots, E_N^+)^{\mathsf{T}}$.

Transfer matrices. Every component maps the fields on its left to the fields on its right,

$$\mathbf{E}_{\text{right}} = M\,\mathbf{E}_{\text{left}},$$

so a circuit of components $B_1, B_2, \ldots, B_n$ (in the order the light meets them) has

$$\boxed{T = B_n \cdots B_2\, B_1} \qquad (2N \times 2N \text{ bidirectional},\ N \times N \text{ forward only}).$$

Local matrices. Each component has a small *local* matrix on the waveguides it touches (2×2 for one waveguide, 4×4 for two, …, bidirectional). It is embedded in the identity: if the local matrix M acts on the index set $\mathcal{I}$, then $B = \mathbb{I}$ with $B[\mathcal{I}, \mathcal{I}] = M$, and every other waveguide passes unchanged. A component placed on "all waveguides" (waveguide section, chip edge) is block-diagonal, one block per waveguide.

Phase sign. Forward propagation over a length L picks up $\mathrm{e}^{-\mathrm{j}\beta L}$; the directional coupler is $\left(\begin{smallmatrix} \tau & -\mathrm{j}\kappa \\ -\mathrm{j}\kappa & \tau \end{smallmatrix}\right)$ in the forward basis. Components have no length of their own unless a length is one of their parameters.

Radiation channels. Every Y-splitter gets an extra lane numbered after the real waveguides ($N+1, N+2, \ldots$). It carries the light the junction radiates away, so the circuit matrix is $2(N + N_Y)$ square and stays lossless; the power in these lanes is reported as *lost* light.

Units. Wavelength λ in nm, lengths L in µm, waveguide loss in dB/cm, Bragg coupling κ_g in mm^{-1}; all are converted to SI (m, m^{-1}) inside the formulas below.

2 Propagation

2.1 Propagation constant $\beta(\lambda)$

Without a group index (the default, constant effective index n_{eff}):

$$\beta(\lambda) = \frac{2\pi\, n_{\text{eff}}}{\lambda}.$$

With a group index n_g and reference wavelength λ_0 (first-order dispersion):

$$\beta(\lambda) = \frac{2\pi\, n_{\text{eff}}}{\lambda_0} + \frac{2\pi\, n_g}{\lambda_0^2}\,(\lambda_0 - \lambda),$$

so n_{eff} fixes the phase at λ_0 and $n_g = n_{\text{eff}} - \lambda\, \mathrm{d}n_{\text{eff}}/\mathrm{d}\lambda$ fixes how fast it changes. With $n_g = n_{\text{eff}}$ this is the tangent line of the constant-index model at λ_0 (equal at λ_0, off by about $2\pi n_{\text{eff}}(\lambda - \lambda_0)^2/\lambda_0^3$ elsewhere). The same $\beta(\lambda)$ is used by every component that has a length.

2.2 Loss

A loss of ℓ dB/cm is the power attenuation coefficient

$$\alpha = \ell \cdot 100 \cdot \frac{\ln 10}{10}\ \ \mathrm{m}^{-1}, \qquad \text{field amplitude} \propto \mathrm{e}^{-\alpha L/2}.$$

A lumped loss of ℓ dB is the field factor $a = 10^{-\ell/20}$.

2.3 One pass of a waveguide

$$\boxed{g(L,\lambda) = \exp\Big[-\Big(\mathrm{j}\beta(\lambda) + \tfrac{\alpha}{2}\Big)L\Big]}$$

Every length-dependent formula is written in terms of g (rings use $e = g(L_{\text{ring}})$, one trip round the ring).

3 Component matrices

Each box gives the local bidirectional matrix (basis E^+, E^- per waveguide) and the forward-only version. For two-waveguide components the basis is $(E_1^+, E_1^-, E_2^+, E_2^-)$.

Waveguide section (length L_k on each waveguide k)

$$W(L) = \begin{pmatrix} g(L) & 0 \\ 0 & 1/g(L) \end{pmatrix}, \qquad B = \operatorname{diag}\big(W(L_1), W(L_2), \ldots, W(L_N)\big).$$

FORWARD ONLY: $\operatorname{diag}\big(g(L_1), \ldots, g(L_N)\big)$.
Two arms differing by ΔL give a Mach–Zehnder free spectral range FSR $\approx \lambda^2/(n_g\,\Delta L)$.

Phase shifter (heater) (extra phase θ, wavelength-independent)

$$P(\theta) = \begin{pmatrix} \mathrm{e}^{-\mathrm{j}\theta} & 0 \\ 0 & \mathrm{e}^{\mathrm{j}\theta} \end{pmatrix} \qquad \text{FORWARD ONLY: } \mathrm{e}^{-\mathrm{j}\theta}.$$

Loss element (loss ℓ dB, $a = 10^{-\ell/20}$)

$$A(\ell) = \begin{pmatrix} a & 0 \\ 0 & 1/a \end{pmatrix} \qquad \text{FORWARD ONLY: } a.$$

Partial mirror

Power-reflectance form ($0 \le R < 1$; reflects $\mathrm{j}\sqrt{R}$ from either side, transmits $t = \sqrt{1-R}$):

$$M_R = \frac{1}{t}\begin{pmatrix} 1 & \mathrm{j}\sqrt{R} \\ -\mathrm{j}\sqrt{R} & 1 \end{pmatrix}.$$

Field-reflection form ($-1 < r < 1$, Fresnel/Stokes: reflects r from the left, $-r$ from the right, $t = \sqrt{1-r^2}$):

$$M_r = \frac{1}{t}\begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}, \qquad M_r\, M_{-r} = \mathbb{I}.$$

FORWARD ONLY: t (what passes) in both forms.

Chip edge / grating coupler (loss ℓ dB, reflection R; on every waveguide)

Loss followed by a mirror, the reflection being the one seen from the input side:

$$E = A(\ell)\, M_R = \frac{1}{t}\begin{pmatrix} a & \mathrm{j}a\sqrt{R} \\ -\mathrm{j}\sqrt{R}/a & 1/a \end{pmatrix}, \quad t = \sqrt{1-R} \qquad \text{FORWARD ONLY: } a\sqrt{1-R}.$$

Sagnac loop mirror (SLM) (coupler κ, $\tau = \sqrt{1-\kappa^2}$, loop phase φ)

$$S(\kappa,\varphi) = \frac{1}{1-2\kappa^2}\begin{pmatrix} \mathrm{e}^{-\mathrm{j}\varphi} & -2\mathrm{j}\kappa\tau \\ 2\mathrm{j}\kappa\tau & \mathrm{e}^{\mathrm{j}\varphi} \end{pmatrix} \qquad \text{FORWARD ONLY: } (1-2\kappa^2)\,\mathrm{e}^{-\mathrm{j}\varphi}.$$

The loop phase is either set directly, or follows the loop length: $\varphi = \beta(\lambda)\, L_{\text{loop}}$. In that case the loop also has the waveguide loss: $\mathrm{e}^{-\mathrm{j}\varphi}$ becomes $g(L_{\text{loop}})$ and $\mathrm{e}^{\mathrm{j}\varphi}$ becomes $1/g(L_{\text{loop}})$, i.e. $\varphi \to \varphi - \mathrm{j}\alpha L_{\text{loop}}/2$. On its own the SLM reflects $|\rho|^2 = 4\kappa^2(1-\kappa^2)$ of the power and passes $(1-2\kappa^2)^2$. At $\kappa^2 = \frac{1}{2}$ it is a perfect mirror and has no transfer matrix (the app refuses this value in the bidirectional model). The "SLM pair" is $\mathrm{diag}(S_1, S_2)$, one on each waveguide.

Directional coupler (field coupling κ, $\tau = \sqrt{1-\kappa^2}$; κ^2 of the power crosses)

$$K(\kappa) = \begin{pmatrix} \tau & 0 & -\mathrm{j}\kappa & 0 \\ 0 & \tau & 0 & \mathrm{j}\kappa \\ -\mathrm{j}\kappa & 0 & \tau & 0 \\ 0 & \mathrm{j}\kappa & 0 & \tau \end{pmatrix} \qquad \text{FORWARD ONLY: } \begin{pmatrix} \tau & -\mathrm{j}\kappa \\ -\mathrm{j}\kappa & \tau \end{pmatrix}.$$

Wavelength-dependent coupler (κ_0 at λ_0, slope s in % per nm)

The coupling angle p grows linearly with wavelength:

$$p(\lambda) = \arcsin(\kappa_0)\left[1 + \frac{s}{100}(\lambda - \lambda_0)\right], \qquad \tau = \cos p, \quad \kappa = \sin p,$$

and the matrix is K above with these τ, κ (forward: the same 2×2).

MMI 50/50 splitter (excess loss ℓ dB, $a = 10^{-\ell/20}$, $s = 1/\sqrt{2}$)

$$\begin{pmatrix} as & 0 & -\mathrm{j}as & 0 \\ 0 & s/a & 0 & \mathrm{j}s/a \\ -\mathrm{j}as & 0 & as & 0 \\ 0 & \mathrm{j}s/a & 0 & s/a \end{pmatrix} \qquad \text{FORWARD ONLY: } as \begin{pmatrix} 1 & -\mathrm{j} \\ -\mathrm{j} & 1 \end{pmatrix}.$$

Waveguide crossing (ideal; swaps two neighbouring waveguides)

$$X = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \qquad \text{FORWARD ONLY: } \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

All-pass ring resonator (coupling κ, $\tau = \sqrt{1-\kappa^2}$, circumference L, backscatter r_s)

One trip round the ring is $e = g(L, \lambda) = a\,\mathrm{e}^{-\mathrm{j}\theta}$. Without backscatter ($r_s = 0$):

$$t = \frac{\tau - e}{1 - \tau e}, \qquad \mathcal{R} = \begin{pmatrix} t & 0 \\ 0 & 1/t \end{pmatrix} \qquad \text{FORWARD ONLY: } t.$$

With backscatter: a lossless point scatterer half-way round couples the clockwise and counter-clockwise waves (transmission $t_s = \sqrt{1 - r_s^2}$, reflection $\mathrm{j}r_s$). With

$$q = 1 - \tau t_s e, \qquad G = t_s e - \frac{r_s^2\,\tau\,e^2}{q} \quad \text{(one trip, including the detour through the other direction)},$$

$$t = \frac{\tau - G}{1 - \tau G}, \qquad \rho = \frac{-\mathrm{j}\kappa^2 r_s\, e}{(1 - \tau G)\, q}, \qquad \mathcal{R} = \frac{1}{t}\begin{pmatrix} t^2 - \rho^2 & \rho \\ -\rho & 1 \end{pmatrix} \qquad \text{FORWARD ONLY: } t.$$

ρ is the reflection back into the bus (the same from either side). Each resonance splits into two, at round-trip phases $2\pi m \pm \arcsin r_s$, i.e. $\Delta\lambda_{\text{split}} = \frac{\arcsin r_s}{\pi}$ FSR. Special case $\kappa = 0$: $t = 1$, $\rho = 0$ (the ring is not coupled, also on an exact resonance).

Resonance spacing: FSR $= \lambda_0^2/(n_g L)$ (with no group index set, $\lambda^2/(n_{\text{eff}} L)$).

Add-drop ring (couplings κ_1 to waveguide 1, κ_2 to waveguide 2, $\tau_i = \sqrt{1-\kappa_i^2}$, circumference L)

With the half and full trips $e_h = g(L/2, \lambda)$, $e = e_h^2$:

$$D = 1 - \tau_1\tau_2 e, \qquad t = \frac{\tau_1 - \tau_2 e}{D}, \qquad t' = \frac{\tau_2 - \tau_1 e}{D}, \qquad d = \frac{-\kappa_1\kappa_2\, e_h}{D}.$$

t, t' are the through transmissions of waveguides 1 and 2 and d the drop. Dropped light leaves the *other* waveguide travelling *backwards* (the drop port is on the input side), which gives

$$\begin{pmatrix} t - d^2/t' & 0 & 0 & d/t' \\ 0 & 1/t & -d/t & 0 \\ 0 & d/t & t' - d^2/t & 0 \\ -d/t' & 0 & 0 & 1/t' \end{pmatrix} \qquad \text{FORWARD ONLY: } \begin{pmatrix} t & 0 \\ 0 & t' \end{pmatrix}.$$

If $\kappa_1 = \kappa_2 = 0$ the matrix is $\mathbb{I}$.

Ring set by a resonance wavelength (any ring: all-pass, add-drop, CROW, OSSB filter)

Instead of its length, a ring can be given a wavelength λ_{res} and a rough length L_0. It then uses the length nearest L_0 that holds a whole number of wavelengths at λ_{res}, so a resonance sits exactly there (also with dispersion):

$$m = \max\bigl(1,\ \text{round}\bigl(\beta(\lambda_{\text{res}})\,L_0/2\pi\bigr)\bigr), \qquad L = \frac{2\pi m}{\beta(\lambda_{\text{res}})}.$$

Coupled-resonator waveguide, CROW (n identical rings of circumference L in a chain between two waveguides)

Ring 1 is beside waveguide 1 and ring n beside waveguide 2. The $n+1$ couplers are $\kappa_0 = \kappa_1$ (waveguide 1 – ring 1), $\kappa_i = \kappa_r$ between neighbouring rings, and $\kappa_n = \kappa_2$ (ring n – waveguide 2); $\tau_i = \sqrt{1-\kappa_i^2}$ and $h = g(L/2, \lambda)$ is half a trip. Looking up the chain from coupler i, Γ_i is what comes back (nothing enters waveguide 2):

$$\Gamma_n = 0, \qquad \Gamma_{i-1} = h^2\,\frac{\tau_i - \Gamma_i}{1 - \tau_i\Gamma_i}, \qquad t = \frac{\tau_0 - \Gamma_0}{1 - \tau_0\Gamma_0},$$

$$x_0 = \frac{-\mathrm{j}\kappa_0}{1 - \tau_0\Gamma_0}, \qquad x_i = \frac{-\mathrm{j}\kappa_i\,h\,x_{i-1}}{1 - \tau_i\Gamma_i}, \qquad d = x_n,$$

and t' is t computed from the other end (the κ_i reversed). With one ring these are the add-drop ring's t, t', d. The rings turn alternately clockwise and counter-clockwise, so with an *odd* n the drop leaves waveguide 2 travelling backwards and the matrix has the add-drop form above; with an *even* n it goes forwards:

$$\Delta = tt' - d^2, \qquad \begin{pmatrix} t & 0 & d & 0 \\ 0 & t'/\Delta & 0 & -d/\Delta \\ d & 0 & t' & 0 \\ 0 & -d/\Delta & 0 & t/\Delta \end{pmatrix} \qquad \text{FORWARD ONLY: } \begin{pmatrix} t & d \\ d & t' \end{pmatrix} \quad \bigl(\text{odd } n\text{: } \operatorname{diag}(t, t')\bigr).$$

Mach–Zehnder interferometer, MZI (couplers κ_1, κ_2; arm a of length L_a with a heater θ, arm b of length L_b)

With $A = \mathrm{e}^{-\mathrm{j}\theta}\,g(L_a, \lambda)$ on waveguide 1 and $B = g(L_b, \lambda)$ on waveguide 2,

$$M = K(\kappa_2)\ \operatorname{diag}\bigl(A,\ 1/A,\ B,\ 1/B\bigr)\,K(\kappa_1),$$

$$\text{FORWARD ONLY: } \begin{pmatrix} \tau_1\tau_2 A - \kappa_1\kappa_2 B & -\mathrm{j}(\tau_2\kappa_1 A + \kappa_2\tau_1 B) \\ -\mathrm{j}(\kappa_2\tau_1 A + \tau_2\kappa_1 B) & \tau_1\tau_2 B - \kappa_1\kappa_2 A \end{pmatrix}.$$

Nothing is reflected. Fringes are $\lambda_0^2/(n_g\,|L_b - L_a|)$ apart.

OSSB filter (ring-assisted MZI) (both couplers κ; on arm a an all-pass ring κ_r, L)

The MZI above with $\kappa_1 = \kappa_2 = \kappa$ and an all-pass ring on arm a: $A = t_r\,\mathrm{e}^{-\mathrm{j}\theta}\,g(L_a, \lambda)$, with $t_r = (\tau_r - e)/(1 - \tau_r e)$, $e = g(L, \lambda)$. Near each resonance the ring's phase turns by 2π over a narrow band, which gives the MZI one very steep edge: one sideband of a modulated carrier passes, the other is blocked. With $L = 2|L_b - L_a|$ and $\theta = \pi/2$ the pass and stop bands alternate every half ring FSR.

Uniform Bragg grating (Bragg wavelength λ_B, length L, coupling κ_g; coupled-mode theory)

The period is chosen so the reflection peaks at λ_B: $\beta_B = \beta(\lambda_B) = \pi/\Lambda$. With the detuning (waveguide loss enters as $\beta \to \beta - \mathrm{j}\alpha/2$)

$$\delta = \beta(\lambda) - \beta_B - \mathrm{j}\frac{\alpha}{2}, \qquad \gamma = \sqrt{\kappa_g^2 - \delta^2}, \qquad C = \cosh\gamma L, \qquad S = \frac{\sinh\gamma L}{\gamma} \quad (S = L \text{ if } \gamma \to 0),$$

$$G_B = \begin{pmatrix} \mathrm{e}^{-\mathrm{j}\beta_B L}\,(C - \mathrm{j}\delta S) & -\mathrm{j}\kappa_g S\,\mathrm{e}^{-\mathrm{j}\beta_B L} \\ \mathrm{j}\kappa_g S\,\mathrm{e}^{\mathrm{j}\beta_B L} & \mathrm{e}^{\mathrm{j}\beta_B L}\,(C + \mathrm{j}\delta S) \end{pmatrix} \qquad \text{FORWARD ONLY: } \frac{\mathrm{e}^{-\mathrm{j}\beta_B L}}{C + \mathrm{j}\delta S}.$$

Lossless peak reflection $R_{\max} = \tanh^2(\kappa_g L)$; first reflection null $\Delta\lambda = \dfrac{\lambda_0^2}{2\pi n_g L}\sqrt{(\kappa_g L)^2 + \pi^2}$ from the peak. With $\kappa_g = 0$ the grating is exactly a waveguide section of length L. The matrix is the solution of the coupled-mode equations $R' = -\mathrm{j}\delta R - \mathrm{j}\kappa_g S$, $S' = \mathrm{j}\kappa_g R + \mathrm{j}\delta S$ for $E^+ = R\,\mathrm{e}^{-\mathrm{j}\beta_B z}$, $E^- = S\,\mathrm{e}^{\mathrm{j}\beta_B z}$, with the teeth counted from the left end ($z = 0$). Two gratings in a row are therefore in step only when the first is a whole number of periods long ($\mathrm{e}^{2\mathrm{j}\beta_B L_1} = 1$, period $\Lambda = \lambda_B/2n_{\text{eff}}$).

Y-splitter (stem on waveguide k, branches on k and $k{+}1$, plus its radiation lane r)

The junction is a real orthogonal matrix U (so it is lossless), used for both directions: the bidirectional local matrix is $U \otimes \mathbb{I}_2$, i.e. $M[2i, 2j] = M[2i{+}1, 2j{+}1] = U_{ij}$. Lane order is (stem, branch, …, radiation); $s = 1/\sqrt{2}$.

Plain junction, stem on the upper waveguide (lanes k, $k{+}1$, r):

$$U_{\text{split}} = \begin{pmatrix} s & 0 & s \\ s & 0 & -s \\ 0 & 1 & 0 \end{pmatrix}, \qquad U_{\text{combine}} = U_{\text{split}}^{\mathsf{T}} = \begin{pmatrix} s & s & 0 \\ 0 & 0 & 1 \\ s & -s & 0 \end{pmatrix}.$$

Splitting, the stem's light goes s into each branch; combining, the in-phase part leaves on the stem and the out-of-phase part, $(E_k - E_{k+1})/\sqrt{2}$, is radiated.

Stem on the lower waveguide:

$$U_{\text{split}} = \begin{pmatrix} 0 & s & -s \\ 0 & s & s \\ 1 & 0 & 0 \end{pmatrix}, \qquad U_{\text{combine}} = \begin{pmatrix} 0 & 0 & 1 \\ s & s & 0 \\ -s & s & 0 \end{pmatrix}.$$

"Make room" (stem on top): the waveguides below the branch shift down one place (split) or up (combine), so the junction acts on m lanes: $k, \ldots, N$ and r. For the split, with $\ell = m - 2$ (indices from 0),

$$U_{00} = U_{0,m-1} = U_{10} = s, \quad U_{1,m-1} = -s, \quad U_{j+1,j} = 1 \ (1 \le j < \ell), \quad U_{m-1,\ell} = 1,$$

all other entries 0; the combine is its transpose. Light already on the bottom waveguide has nowhere to move to and goes into the radiation lane (the app warns about this).

Path lengths: an optional branch length L_b adds $W(L_b)$ on both branches (after a split, before a combine), and a shift length L_s adds $W(L_s)$ on every other waveguide that steps half a spacing aside at the junction. With $L_s = L_b$ all paths gain the same phase.

FORWARD ONLY: the same U (it has no reflections).

Custom components (groups)

A saved sub-circuit is opened up: its components are placed, in order, on the waveguides it occupies (offset by its position), so it contributes the product of its own component matrices. Each Y-splitter inside still gets its own radiation lane.

4 Forward-only model

The forward-only matrix of a component is its forward transmission with nothing entering from the right and the reflections discarded. Splitting a local bidirectional matrix into forward (f) and backward (b) rows and columns, it is the Schur complement

$$\boxed{M_{\text{fwd}} = M_{ff} - M_{fb}\, M_{bb}^{-1}\, M_{bf}} \qquad \big(= 1/M_{22} \text{ for one waveguide}\big),$$

and $M_{\text{fwd}} = M_{ff}$ when the component does not reflect. The circuit is then $T = B_n \cdots B_1$ with these $N \times N$ matrices. (Note: the forward-only model multiplies the per-component transmissions, so it ignores multiple reflections *between* components; the bidirectional model includes them.)

5 Solving the circuit

5.1 Inputs

Light enters on the left only. The input amplitude vector $\mathbf{A}$ is one of

$$\mathbf{A} = \mathbf{e}_k \text{ (one waveguide)}, \qquad A_k = 1\ \forall k \text{ (all, in phase)}, \qquad A_k = a_k\, \mathrm{e}^{\mathrm{j}\phi_k} \text{ (custom)},$$

padded with zeros on the radiation lanes. The input power is $P_{\text{in}} = \sum_k |A_k|^2$.

5.2 Forward-only

$$\mathbf{o} = T\,\mathbf{A}, \qquad \mathbf{r} = \mathbf{0}.$$

5.3 Bidirectional

On the left the forward fields are known ($\mathbf{A}$) and the backward fields $\mathbf{b}$ (the reflections) are unknown; on the right nothing comes back in, $\mathbf{E}^{-}_{\text{right}} = \mathbf{0}$. Splitting T into forward/backward blocks:

$$\begin{pmatrix} \mathbf{o} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} T_{ff} & T_{fb} \\ T_{bf} & T_{bb} \end{pmatrix} \begin{pmatrix} \mathbf{A} \\ \mathbf{b} \end{pmatrix} \quad \Longrightarrow \quad \boxed{\mathbf{b} = -T_{bb}^{-1}\, T_{bf}\, \mathbf{A}, \qquad \mathbf{o} = T_{ff}\, \mathbf{A} + T_{fb}\, \mathbf{b}}$$

(T_{bb} is solved by Gaussian elimination with partial pivoting; it is singular only at a genuine blow-up, e.g. a perfect mirror.)

5.4 Reported quantities

$$\begin{aligned}
\text{output power at waveguide } k: &\quad P_k^{\text{out}} = |o_k|^2/P_{\text{in}}, \qquad \text{reflected: } P_k^{\text{back}} = |b_k|^2/P_{\text{in}}, \\
\text{lost at Y-splitter } j: &\quad P_j^{\text{lost}} = \big(|o_{N+j}|^2 + |b_{N+j}|^2\big)/P_{\text{in}}, \\
\text{in dB:} &\quad 10\log_{10}\max(P,\, 10^{-30}), \qquad \text{phase: } \arg o_k, \\
\text{energy balance:} &\quad \mathcal{E} = \textstyle\sum_k P_k^{\text{out}} + \sum_k P_k^{\text{back}} + \sum_j P_j^{\text{lost}} \quad (= 1 \text{ for a lossless circuit}), \\
\text{forward-only:} &\quad \text{“gets through”} = \textstyle\sum_k P_k^{\text{out}}.
\end{aligned}$$

The Matrices tab also shows $|\det T|$, which is exactly 1 for any lossless bidirectional circuit (every component above has $|\det| = 1$ when lossless).

5.5 Spectrum sweep

The spectrum evaluates the whole circuit at N_p wavelengths spread evenly over the span Δ about the centre λ_c:

$$\lambda_i = \lambda_c - \frac{\Delta}{2} + \Delta\,\frac{i}{N_p - 1}, \qquad i = 0, \ldots, N_p - 1,$$

rebuilding $T(\lambda_i)$ and solving for $\mathbf{o}, \mathbf{b}$ at each one. N_p is at most 5001, and a large circuit is given fewer points so the page stays quick: with W an estimate of the work for one wavelength (a fixed part, a part per component, $k^2 D$ for each $k \times k$ component matrix applied to the $D \times D$ circuit matrix, and the solve for $\mathbf{b}$), $N_p \le \max(11,\ \lfloor 4 \times 10^6 / W \rfloor)$. The estimate depends only on the circuit, so the same circuit always gets the same number of points, and the exported Python uses that number.

5.6 Phase and group delay

The Phase chart is the phase $\varphi_k(\lambda) = \arg o_k$ (or $\arg b_k$), followed continuously from its value at the centre wavelength: $\varphi(\lambda_{i+1}) = \varphi(\lambda_i) + \arg\big(o(\lambda_{i+1})\, o^*(\lambda_i)\big)$. The Delay chart is the group delay

$$\boxed{\tau_g = -\frac{\mathrm{d}\varphi}{\mathrm{d}\omega} = \frac{\lambda^2}{2\pi c}\,\frac{\mathrm{d}\varphi}{\mathrm{d}\lambda}}$$

with the derivative taken by central differences on the sweep grid. For a waveguide $\varphi = -\beta L$, so $\tau_g = n_g L/c$ ($n_{\mathrm{eff}} L/c$ with no group index). A lossless all-pass ring on resonance gives $\tau_g = \dfrac{n_g L}{c}\,\dfrac{1+\tau}{1-\tau}$. Where the light at a port passes through zero its phase jumps by π; such a jump is not a delay and is left out.

5.7 Light inside the circuit (Light view and Probe)

With $\mathbf{E}_0 = (A_1, b_1, A_2, b_2, \dots)^{\mathsf{T}}$ the solved field at the input end, the field after the first m components is

$$\mathbf{E}_m = B_m \cdots B_1\, \mathbf{E}_0,$$

and the power travelling forward and backward in waveguide k at that point is $|E^{+}_{m,k}|^2/P_{\mathrm{in}}$ and $|E^{-}_{m,k}|^2/P_{\mathrm{in}}$. The Light view shows these between components. Inside a resonator it shows the circulating power, from the fields arriving at it (a forward from the left, a' backward from the right):

$$\text{all-pass ring:}\quad \begin{pmatrix} q & -\mathrm{j}\tau r_s e \\ -\mathrm{j}\tau r_s e & q \end{pmatrix}\begin{pmatrix} c \\ d \end{pmatrix} = -\mathrm{j}\kappa \begin{pmatrix} a \\ a' \end{pmatrix}, \qquad q = 1 - \tau t_s e \quad \Big(r_s = 0:\ c = \frac{-\mathrm{j}\kappa\, a}{1-\tau e}\Big),$$

$$\text{add-drop ring:}\quad c = \frac{-\mathrm{j}\kappa_1 a_1 - \mathrm{j}\kappa_2 \tau_1 e_h\, a_2'}{1 - \tau_1\tau_2 e}, \qquad d = \frac{-\mathrm{j}\kappa_1 a_1' - \mathrm{j}\kappa_2 \tau_1 e_h\, a_2}{1 - \tau_1\tau_2 e},$$

$$\text{SLM loop:}\quad c = \tau a - \mathrm{j}\kappa a', \qquad d = -\mathrm{j}\kappa a + \tau a',$$

$$\text{OSSB ring:}\quad \text{the all-pass ring, with } a = \tau a_1 - \mathrm{j}\kappa a_2, \quad a' = \mathrm{e}^{-\mathrm{j}\theta} g(L_a)\,(\tau b_1 - \mathrm{j}\kappa b_2),$$

where c, d are the two directions round the ring or loop (b_1, b_2: backward fields arriving on the right). In a CROW the waves in every ring are found together, by solving the coupler equations for all rings at once. A lossless ring of coupling κ on resonance holds $|c|^2 = \kappa^2/(1-\tau)^2$ times the power arriving; an add-drop ring with $\kappa_1 = \kappa_2 = \kappa$ holds $1/\kappa^2$ times.

6 Derived results used as checks

The Python library runs 64 self-checks (`python opticsbox.py`). They include the closed-form results below, and independent derivations: the Sagnac loop mirror, the ring (with loss and backscatter), the add-drop ring and the CROW (1 to 5 rings) are each rebuilt from couplers, lengths of waveguide and point reflectors and solved for their internal waves; the Bragg grating is compared with a thin-film stack of 600 quarter-wave periods; every component's scattering matrix is checked to be symmetric (reciprocal), and unitary when lossless; and 200 random circuits are solved a second way, as a network of scattering matrices.

Coupler split	κ^2 crosses, $1-\kappa^2$ stays.
Mach–Zehnder	FSR $\approx \lambda^2/(n_g\,\Delta L)$.
Ring / add-drop ring	FSR $= \lambda_0^2/(n_g L)$ (no group index: $\lambda^2/(n_{\text{eff}} L)$).
Ring backscatter splitting	$\Delta\lambda = (\arcsin r_s/\pi)\,\text{FSR}$.
SLM alone	reflects $4\kappa^2(1-\kappa^2)$, passes $(1-2\kappa^2)^2$.
Bragg grating	$R_{\max} = \tanh^2(\kappa_g L)$; first null at $\frac{\lambda_0^2}{2\pi n_g L}\sqrt{(\kappa_g L)^2+\pi^2}$.
Fabry–Pérot (two mirrors, gap L)	$T = \dfrac{(1-R_1)(1-R_2)\,a^2}{(1-\sqrt{R_1R_2}\,a^2)^2 + 4\sqrt{R_1R_2}\,a^2\sin^2(\delta/2)}$, with $a^2 = \mathrm{e}^{-\alpha L}$ and round-trip phase $\delta = 2\beta L - \arg(\rho_1\rho_2)$ (ρ_i: mirror reflections seen from inside the cavity); FSR $= \lambda_0^2/(2n_g L)$.
Group delay	waveguide $n_g L/c$; all-pass ring on resonance $\frac{n_g L}{c}\frac{1+\tau}{1-\tau}$.
Lossless circuits	$\mathcal{E} = 1$ and $\lvert\det T\rvert = 1$; every component reciprocal ($S = S^{\mathsf{T}}$).